

Inversion Algorithm (class lecture)

Thursday, April 9, 2026 12:10 PM

Inverse of a matrix

Reduce echelon form

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Through Row operation

$$A A^{-1} = I$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj } A$$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$|A| = ad - cb$$

$$\text{adj } A = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$A^{-1} = \frac{1}{ad - cb} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Let $A = \begin{pmatrix} 3 & 2 \\ -3 & -1 \end{pmatrix}$

$$|A| = -3 + 6 = 3$$

$$\text{adj } A = \begin{pmatrix} -1 & -2 \\ 3 & 3 \end{pmatrix}$$

$$A^{-1} = \frac{1}{3} \begin{pmatrix} -1 & -2 \\ 3 & 3 \end{pmatrix} = \begin{pmatrix} -1/3 & -2/3 \\ 1 & 1 \end{pmatrix}$$

Inversion Algorithm

$$\begin{pmatrix} 3 & 2 & | & 1 & 0 \\ -3 & -1 & | & 0 & 1 \end{pmatrix} \xrightarrow{\frac{1}{3}R_1} \begin{pmatrix} 1 & \frac{2}{3} & | & \frac{1}{3} & 0 \\ -3 & -1 & | & 0 & 1 \end{pmatrix}$$

$$R_2 + 3R_1 \rightarrow \begin{pmatrix} 1 & \frac{2}{3} & | & \frac{1}{3} & 0 \\ 0 & 1 & | & 1 & 1 \end{pmatrix}$$

$$R_1 - \frac{2}{3}R_2 \rightarrow \begin{pmatrix} 1 & 0 & | & -\frac{1}{3} & -\frac{2}{3} \\ 0 & 1 & | & 1 & 1 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} -\frac{1}{3} & -\frac{2}{3} \\ 1 & 1 \end{pmatrix}$$

$$\begin{array}{l} R_2 - \frac{1}{3}R_1 \rightarrow \begin{pmatrix} 1 & \frac{2}{3} & | & \frac{1}{3} & 0 \\ 0 & -1 & | & -\frac{1}{3} & 1 \end{pmatrix} \\ 3R_2 \rightarrow \begin{pmatrix} 1 & \frac{2}{3} & | & \frac{1}{3} & 0 \\ 0 & -3 & | & -1 & 3 \end{pmatrix} \\ R_1 + \frac{2}{3}R_2 \rightarrow \begin{pmatrix} 1 & 0 & | & -\frac{1}{3} & -\frac{2}{3} \\ 0 & -3 & | & -1 & 3 \end{pmatrix} \\ R_1 - \frac{2}{3}R_2 \rightarrow \begin{pmatrix} 1 & 0 & | & -\frac{1}{3} & -\frac{2}{3} \\ 0 & -3 & | & -1 & 3 \end{pmatrix} \end{array}$$

$$\vec{A} = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 3 & -1 & 2 & | & 1 & 0 & 0 \\ 2 & 4 & -4 & | & 0 & 1 & 0 \\ 5 & -1 & -3 & | & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{array}{r} R_2: 2 \quad 4 \quad -4 \quad 0 \quad 1 \quad 0 \\ 2R_1: 3 \quad -1 \quad 2 \quad 0 \quad 0 \quad 1 \\ \hline 0 \quad 14 \quad -16 \quad -2 \quad 3 \quad 0 \end{array}$$

$$R_1 - R_2 \begin{pmatrix} 1 & -5 & 6 & | & 1 & -1 & 0 \\ 2 & 4 & -4 & | & 0 & 1 & 0 \\ 5 & -1 & -3 & | & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{array}{r} R_3: 5 \quad -1 \quad -3 \quad 0 \quad 0 \quad 1 \\ 5R_1: 5 \quad -25 \quad -30 \quad -5 \quad -5 \quad 5 \\ \hline 0 \quad 24 \quad -33 \quad -5 \quad 5 \quad 0 \end{array}$$

$$\begin{array}{l} R_2 - 2R_1 \\ R_3 - 5R_1 \end{array} \begin{pmatrix} 1 & -5 & 6 & | & 1 & -1 & 0 \\ 0 & 14 & -16 & | & -2 & 3 & 0 \\ 0 & 24 & -33 & | & -5 & 5 & 1 \end{pmatrix}$$

$$\frac{1}{14} R_2 \begin{pmatrix} 1 & -5 & 6 & | & 1 & -1 & 0 \\ 0 & 1 & -\frac{16}{14} & | & -\frac{2}{14} & \frac{3}{14} & 0 \\ 0 & 24 & -33 & | & -5 & 5 & 1 \end{pmatrix} \quad -1 + \frac{15}{14}$$

$$\begin{array}{l} R_1 + 5R_2 \\ R_3 - 24R_2 \end{array} \begin{pmatrix} 1 & 0 & \frac{4}{14} & | & \frac{4}{14} & \frac{1}{14} & 0 \\ 0 & 1 & -\frac{16}{14} & | & -\frac{2}{14} & \frac{3}{14} & 0 \\ 0 & 0 & -\frac{78}{14} & | & \frac{22}{14} & -\frac{2}{14} & 1 \end{pmatrix}$$

$$\begin{array}{r} R_3: 0 \quad 24 \quad -33 \quad -5 \quad 5 \quad 1 \\ 24R_2: 0 \quad 24 \quad -384 \quad -48 \quad 72 \quad 0 \\ \hline 0 \quad 0 \quad -78 \quad \frac{22}{14} \quad -\frac{2}{14} \quad 1 \end{array}$$

Ty Ty Ty '

$$\begin{array}{l}
 -\frac{14}{78}R_3 \\
 R_1 - \frac{4}{14}R_3 \\
 R_2 + \frac{16}{14}R_3
 \end{array}
 \left(
 \begin{array}{ccc|ccc}
 1 & 0 & \frac{4}{14} & \frac{4}{14} & \frac{1}{14} & 0 \\
 0 & 1 & -\frac{16}{14} & -\frac{2}{14} & \frac{3}{14} & 0 \\
 0 & 0 & 1 & -\frac{22}{78} & \frac{2}{78} & -\frac{14}{78}
 \end{array}
 \right)
 \begin{array}{l}
 -\frac{22}{14} \times \frac{14}{78} \\
 -\frac{2}{14} \times \frac{14}{78}
 \end{array}$$

$$\left(
 \begin{array}{ccc|ccc}
 & & & & & \\
 & & & & & \\
 0 & 0 & 1 & -\frac{22}{78} & \frac{2}{78} & -\frac{14}{78}
 \end{array}
 \right)$$